

Bringing LAMBDA to Excel

Jack Williams - Microsoft Research

aka.ms/CalcIntel

Date	Name	Symbol
04/06/2023	Flight out	✈️
05/06/2023	LAMBDA Days	λ
06/06/2023	LAMBDA Days	λ
07/06/2023	Flight back	✈️
25/12/2023	Christmas	🎄

Excel Labs

Grid

Names

Modules

📁

Sheet1 > C27

📌

1

ExcelApi 1.17 (PC/16.0.16505.20000)



The Microsoft Garage
@MSFTGarage

What began as [#MicrosoftHackathon](#) idea is now accelerating calculations & bringing client-side estimation to the [#Excel](#) web experience. [@MSFTGarage](#) Wall of Fame inductee Calc.ts in Excel for the web tells their story in their own words aka.ms/AA8xlb1

	A	B	C	D	
1	Year	Category	Sales	Product	Q1 Dis
2	2017	Components	\$ 20,000	Chains	
3	2015	Clothing	\$ 3,700	Socks	
4	2017	Clothing	\$ 4,000	Bib-Shorts	
5	2015	Clothing	\$ 13,300	Shorts	
6	2017	Clothing	\$ 36,000	Tights	
7	2015	Components	\$ 2,300	Handlebars	
8	2016	Clothing	\$ 2,300	Socks	
9	2016	Components	\$ 3,400	Brakes	
10	2016	Bikes	\$ 6,300	Mountain Bikes	
11	2017	Components	\$ 5,400	Brakes	
12	2016	Accessories	\$ 17,000	Helmets	
13	2016	Accessories	\$ 21,600	Lights	
14	2016	Components	\$ 1,000	Bottom Brackets	
15	2015	Clothing	\$ 6,700	Jerseys	
16	2015	Bikes	\$ 3,500	Road Bikes	
17	2017	Clothing	\$ 7,500	Jerseys	
18	2017	Bikes	\$ 9,300	Cargo Bike	
19	2017	Bikes	\$ 8,500	Mountain Bikes	
20	2017	Accessories	\$ 33,700	Bike Racks	
21		Total	=SUM(C2:C20)		Highest
22			SUM(C2:C20)		
23					
24					
25					

Project Yellow

Excel as a Programming Language

[illegible]



Announcing LAMBDA: Turn Excel formulas into custom functions

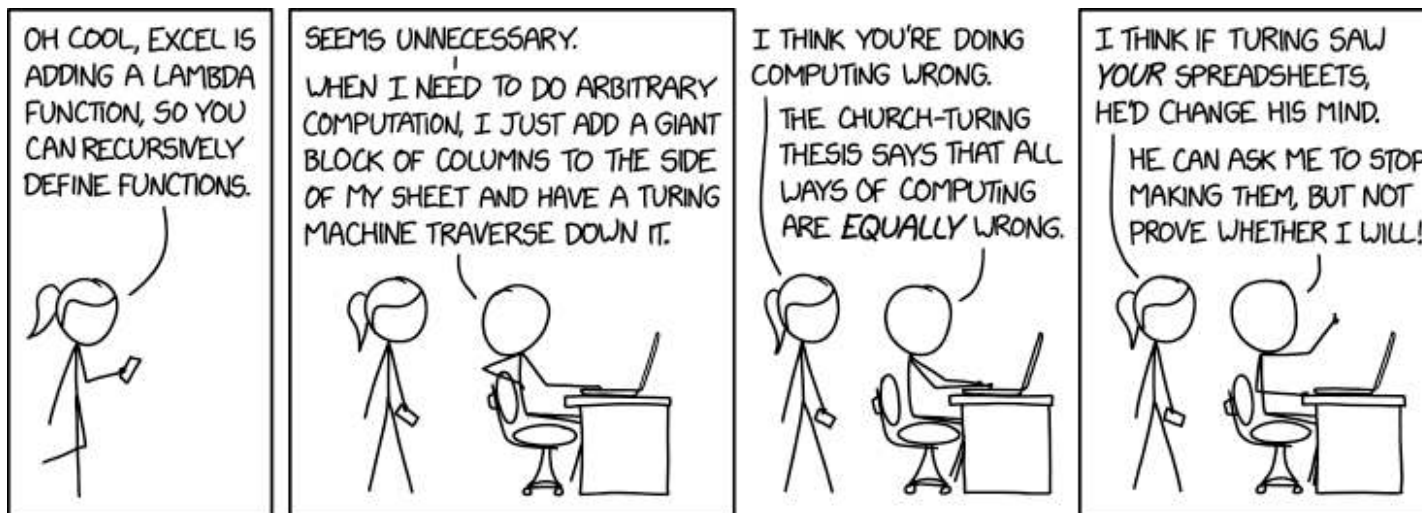
By  [Brian Jones \(OFFICE\)](#)

Published Dec 03 2020 08:00 AM

👁 416K Views

```
=LET(  
    twice, LAMBDA(f, LAMBDA(x, f(f(x)))),  
    plusOne, LAMBDA(x, x + 1),  
    plusFour, twice(twice(plusOne)),  
    MAP(SEQUENCE(5), plusFour)  
)
```

5
6
7
8
9



"[...] In the hands of Excel cowboys, this can become another foot gun."

"I don't think name manager cuts it, they'll need a formula manager. Comments between coding lines and tooltips are going to be necessary. Sharing, vital [...]"

[End-user encounters with lambda abstraction in spreadsheets: Apollo's bow or Achilles' heel? \(VL/HCC 2022\)](#)

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By  [Chris Gross](#)

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 **The Microsoft Garage** @MSFTGarage · 8 Mar 2022

Learn more about a supercharged way to author LAMBDA functions in Excel with support for sharing and more, with Advanced Formula Environment, a @MSFTGarage project [aka.ms/GarageAFEBlog](#)

[#Excel](#) [#productivity](#)



$\Lambda\lambda$
= LAMBDA
+

Advanced Formula Environment

custom functions without code

$(x, y, x+y) \rightarrow 4$

known as *calculus* *mathematical*

a new way to author LAMBDA functions



Excel Labs

GridNamesModules

Function name

GROUPBY

Description

Groups `tbl` by `key\_col` and applies `agg` to each group

Arguments

tblkey_colagg

Function definition

```
=LET(
  positions_for_key, LAMBDA(key,
    FILTER(SEQUENCE(ROWS(key_col)), key_col = key)
  ),
  keys, UNIQUE(key_col),
  reduce_group, LAMBDA(key,
    agg(CHOOSEROWS(tbl, positions_for_key(key)))
  ),
  HSTACK(keys, BYROW(keys, reduce_group))
)
```

DoneCancel

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Sheet4 > G2

1

=GROUPBY(Table2[Sales], Table2[Product], LAMBDA(x, SUM(x)))

	A	B	C	D	E	F	G	H
1	Year	Category	Product	Sales	Rating			
2	2017	Components	Chains	£ 20,000	75%		Chains	45100
3	2015	Clothing	Socks	£ 3,700	22%		Socks	9700
4	2017	Clothing	Bib-Shorts	£ 4,000	22%		Bib-Shorts	7600
5	2015	Clothing	Shorts	£ 13,300	56%		Shorts	48300
6	2017	Clothing	Tights	£ 36,000	100%		Tights	61400
7	2015	Components	Handlebars	£ 2,300	35%		Handlebars	10600
8	2016	Clothing	Socks	£ 2,300	28%		Brakes	11100
9	2016	Components	Brakes	£ 3,400	36%		Mountain Bikes	17900
10	2016	Bikes	Mountain Bikes	£ 6,300	40%		Helmets	59300

Excel Labs

Grid Names Modules



Workbook calendar GENAI + New

```
1 days = {"Mon", "Tue", "Wed", "Thur", "Fri", "Sat", "Sun"};
2
3 /** Returns a row vector with the month name in the first position, and blanks for
4 monthHeader = LAMBDA(serial, HSTACK(TEXT(serial, "mmm"), MAKEARRAY(1, 6, LAMBDA(_i,
5
6 /**
7 * Create a calendar for the month containing the input
8 * day serial.
9 */
10 monthCalendar = LAMBDA(serial, dates, [fmt],
11 LET(
12     fmt, IF(ISOMITTED(fmt), "d", fmt),
13     daysInMonth, EOMONTH(serial, 0) - EOMONTH(serial, -1),
14     foMonth, DATE(YEAR(serial), MONTH(serial), 1),
15     dayPadding, WEEKDAY(foMonth, 2) - 1,
16     calendarRows, ROUNDUP((dayPadding + daysInMonth) / 7, 0),
17     // i, j is the position in the calendar array
18     builder, LAMBDA(i, j,
19         LET(
20             seqNum, ((i - 1) * 7 + j) - dayPadding,
21             currentDate, DATE(YEAR(serial), MONTH(serial), seqNum),
22             matches, FILTER(dates, CHOOSECOLS(dates, 1) = currentDate),
23             matchedContent, CHOOSECOLS(matches, 3),
24             content, IFERROR(ARRAYTOTEXT(FILTER(matchedContent, matchedContent
25             IFS(
26                 // The days before the start of the month
27                 seqNum <= 0,
28                 "",
29                 seqNum <= daysInMonth,
```

1 =calendar.monthCalendar

days

monthCalendar

monthHeader

GitHub Gist

Search...

All gists

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jack-williams / calendar

Secret

Edit

Last active 2 minutes ago

<> Code

Revisions 7

calendar

```
1 days = {"Mon", "Tue", "Wed", "Thur", "Fri", "Sat", "Sun"};
2
3 /** Returns a row vector with the month name in the first position, and blanks for the rest */
4 monthHeader = LAMBDA(serial, HSTACK(TEXT(serial, "mmm"), MAKEARRAY(1, 6, LAMBDA(_i, _j, ""))));
5
6 /**
```

	B	C	D	E	F	G	H	I	J
1									
2		01/12/2023							
3									
4									
5									
6									
7									
8									
9									
10									
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21									
22									
23									

Excel Labs

GridNamesModules





FunctionsRangesFormulasModules

Functions defined in the name manager will appear here. Use the + button to define a new named function.

	A	B	C	D	E	F	G	
1								
2		Fruit	Quantity	Remove Fruit	Remove Paren	Cost	Amount	
3		Apples (0.7)	4	0.7)	0.7	£ 0.70	£ 2.80	
4		Pears (0.9)	2	0.9)	0.9	£ 0.90	£ 1.80	
5		Oranges (1.1)	6	1.1)	1.1	£ 1.10	£ 6.60	
6								
7				=TEXTAFTER(B3, "(")	=TEXTBEFORE(D3, ")")	=VALUE(E3)	=C3*F3	
8								

	B	C	D	E	F	G
1						
2	Fruit	Quantity	Remove Fruit	Remove Paren	Cost	Amount
3	Apples (0.7)	4	0.7)	0.7	£ 0.70	£ 2.80
4	Pears (0.0)	2	0.0)	0.0	£ 0.00	£ 1.00

Excel Labs

Import from grid

Extract formulas from the grid into a function, by pulling out a calculation split into steps across several cells into a reusable function. The input cells will become parameters to the function.

Range containing the calculation

Cells to treat as parameters

Output cell

Cancel

Preview →

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GridNamesModules

Import from grid

Function name

Refers to

```
=LAMBDA(Fruit, Quantity,
    LET(
        Remove_Fruit, TEXTAFTER(Fruit, "("),
        Remove_Paren, TEXTBEFORE(Remove_Fruit, ")"),
        Cost, VALUE(Remove_Paren),
        Quantity * Cost
    )
)
```

← Back

Create

ExcelApi 1.17 (PC/16.0.16505.20000)

A User-Centred Approach to Functions in Excel

30th June 2003

Simon Peyton Jones
Microsoft Research, Cambridge
simonpj@microsoft.com

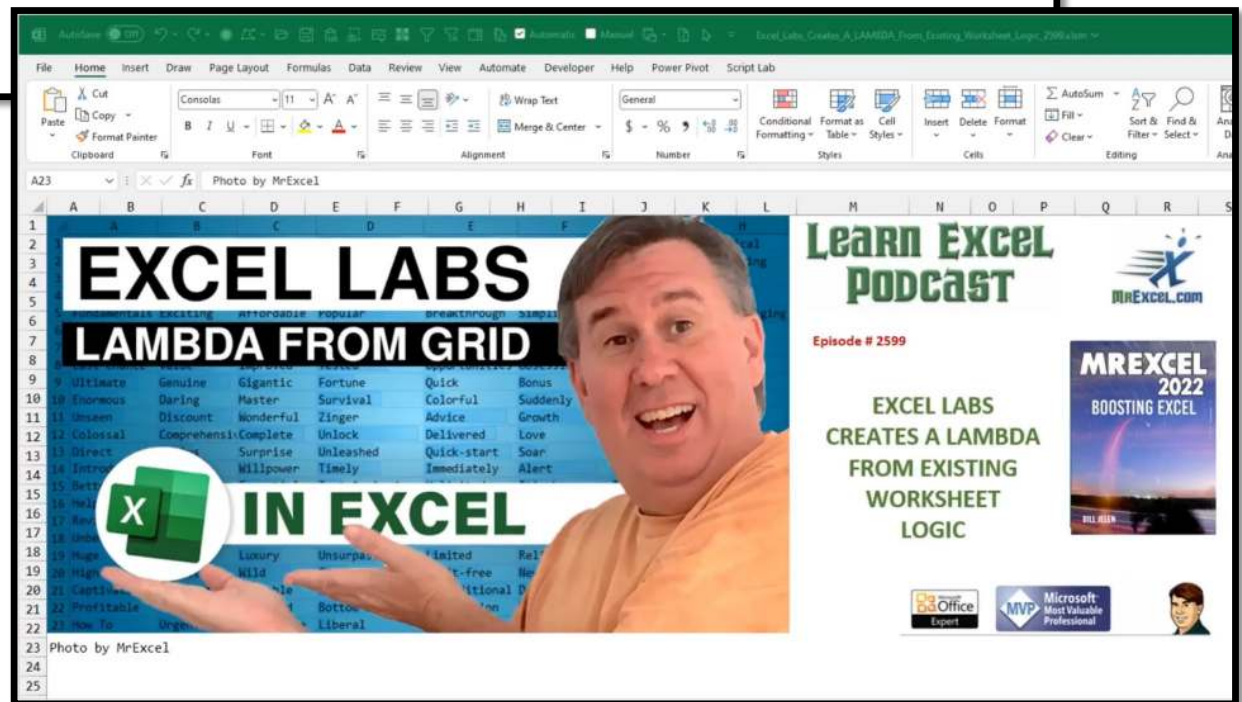
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MrExcel.com
140K subscribers

<https://www.youtube.com/watch?v=9EXERguX90>



Excel Labs, Create A LAMBDA From Existing Worksheet Logic, 2006/6/30

File Home Insert Draw Page Layout Formulas Data Review View Automate Developer Help Power Pivot Script Lab

Clipboard Font Alignment Number Styles Cells Editing

A23 Photo by MrExcel

EXCEL LABS
LAMBDA FROM GRID
IN EXCEL

Learn Excel Podcast
Episode # 2599
EXCEL LABS
CREATES A LAMBDA
FROM EXISTING
WORKSHEET
LOGIC

MREXCEL 2022
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Office Expert MVP Microsoft Most Valuable Professional

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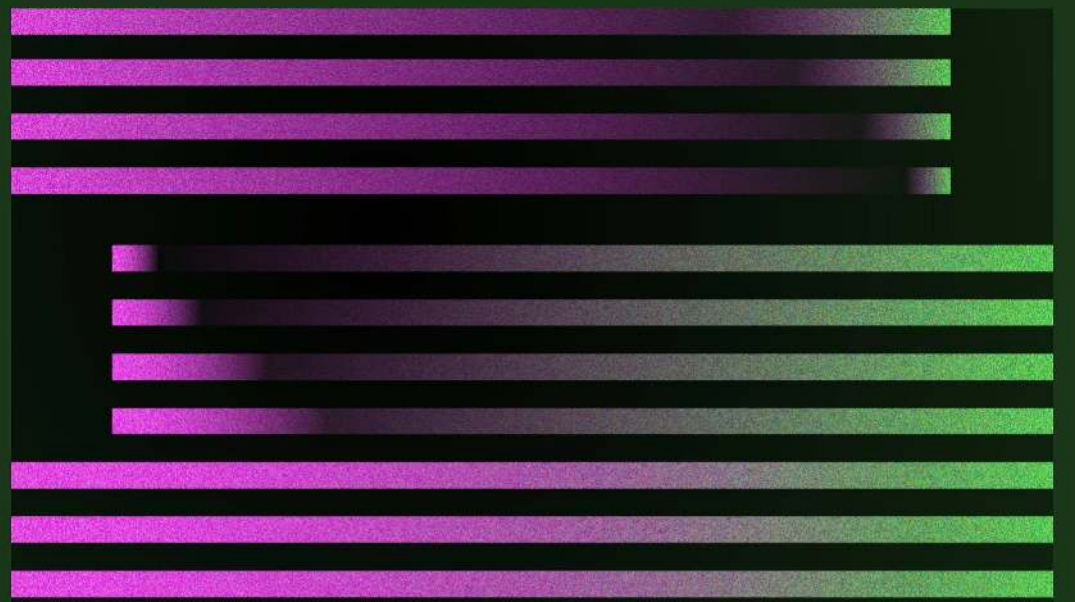
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Experimentation is key to unlocking innovation in any field. It's how we learn and grow, and it's what helps us continue to create features that help you do more.

So today, we are thrilled to announce our continued investment in experimentation through the release of **Excel Labs**, a [Microsoft Garage project](#). Excel Labs is an add-in that allows us to release experimental ideas for you to try, and to give us feedback that helps us evolve Excel to be most useful for you.

Introducing ChatGPT

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests.



[Introducing ChatGPT \(openai.com\)](https://openai.com)

```
=LABS.GENERATIVEAI("What is Lambda days?")
```

Lambda Days is an annual conference focused on functional programming that takes place in Krakow, Poland. It brings together experts and enthusiasts from academia and industry to share their knowledge and experience in the field of functional programming.

Excel Labs

Grid Names Modules

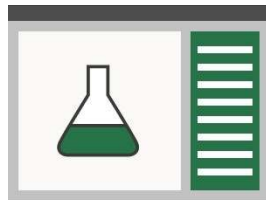


Workbook calendar GENAI + New

```
1  /**
2   * Given a list of categories and an item, return the category that best matches.
3   */
4  CLASSIFY = LAMBDA(categories, item,
5      LABS.GENERATIVEAI(
6          "Given this list of potential categories: " & TEXTJOIN(", ", TRUE, categories) &
7          ", which category best applies to the following item: " & item &
8          ". Your response should be one of the categories (matching exact spelling and case) and nothing else.",
9          0
10     )
11 );
12
13 /**
14 * Standardize the representation of `value` using the described `type`.
15 */
16 STANDARDIZE = LAMBDA(type, value,
17     LABS.GENERATIVEAI(
18         "Given a type and value, standarize the representation of the value using the provided type description." &
19         "The type is: " & type & ". The value is: " & value &
20         ". Return the result exactly. The standardized value is: "
21     )
22 );
23
```

Travel
Holiday
Conference

Date	Name	Symbol	Type
04/06/2023	Flight out		
05/06/2023	LAMBDA Days	λ	
06/06/2023	LAMBDA Days	λ	
07/06/2023	Flight back		
25/12/2023	Christmas		



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